RAJALAKSHMI ENGINEERING COLLEGE

Janarthan R 250701261-2025-CSE

J2

Started on	Tuesday, 25 August 2026, 12:04 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 12:10 PM
Time taken	5 mins 27 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
#include<stdbool.h>

bool binary_search(int n,int arr[],int x){
    int start = 0; int end = n - 1;
    while(start <= end){
        int mid = start + (end - start)/2;
        if(arr[mid] == x){
            return true;
        }

        else if(arr[mid] > x){
            end = mid - 1;
        }
        else start = mid + 1;
    }
    return false;
}

bool function(int arr[],int n,int x){
    for(int i=0;i<n;i++){
        int first = arr[i];
        int second = x - first;
        if(binary_search(n,arr,second)){
            printf("%d\n%d\n",first,second);
            return true;
        }
    }
    return false;
}

int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++) scanf("%d",&arr[i]);
    int x;
    scanf("%d",&x);
    if(!function(arr,n,x)){
        printf("No\n");
        return 0;
    }
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course